

Project 3

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1 Introduction

One of my favorite websites to visit is the Online Encyclopedia of Integer Sequences (OEIS) [3]. It is a vast database of integer sequences that people from all over the world can contribute to and use for research. Each sequence has a unique identifier, starting with the letter ‘A’ followed by six digits. For example, the sequence of prime numbers is identified as A000040. Today, I want to explore the very first sequence in the database: A000001. This sequence counts the number of groups of order n . I will not go into much detail about what a group is, other than it is a set of elements with an operation that satisfies certain properties. The order of a group is just the number of elements in the group. Let us take a look at the first few terms of the sequence A000001:

n	0	1	2	3	4	5	6	7	8	9	10
$a(n)$	0	1	1	1	2	1	2	1	5	2	2

The sequence doesn’t look like it’s growing very quickly at first glance. However, as we look at larger values of n , we can see that the number of groups of order n does indeed increase. For instance, there are 267 groups of order 64. So how fast does the sequence grow? This is a surprisingly difficult question, and to give you a sense of the complexity, how about we play a little game by Matthew Macauley [2]. Here are the values of the number of groups of order n for n from 1014 to 1023:

n	1013	1014	1015	1016	1017	1018	1019	1020	1021	1022	1023
$a(n)$	1	23	2	12	2	2	1	37	1	4	2

Can you guess the number of groups of order 1024? Take a moment to think about it before scrolling down for the answer.

2 Groups of Order 2^n Dominates Conjecture

So the number of groups of order 1024 is 49,487,367,289 which was only determined in 2021 by David Burrell [1]. That's a huge jump from the previous values! One of the reasons why there are so many groups of order 1024 is that 1024 is a power of 2. Without going into too much detail, groups of order p^n for a prime number p tend to have more complex structures and thus more distinct groups. For example, $n = 729 = 3^6$ has 504 groups. However, powers of 2 are so dominant that they account for more than 99% of all groups of order less than 2000. Calculating the exact number of groups of order n as n grows is still very difficult, with John Conway saying that the “human race will never know the exact number of groups of order 2048.” Which brings us to the main conjecture I want to discuss today:

Conjecture Groups of Order 2^n Dominates. Almost all finite groups are of order 2^n . That is

$$\lim_{n \rightarrow \infty} \frac{a(2^n)}{\sum_{k=1}^n a(k)} = 1.$$

Given the information we have so far, this conjecture seems almost trivially true. However, it has not been proven rigorously yet.

3 Bibliography

References

- [1] David Burrell. On the number of groups of order 1024. *Communications in Algebra*, 50(6):2408–2410, 2022.
- [2] Matthew Macauley. Visual algebra, lecture 2.12: Other finite groups, 2025.
- [3] Neil JA Sloane et al. The on-line encyclopedia of integer sequences, 2003.